

Introduction to computational physics Lecture 1

PHYS 246 class 1

Fall 2026

J Noronha-Hostler

[https://jnoronhahostler.github.io/IntroductionToComputationalPhysics/
intro.html](https://jnoronhahostler.github.io/IntroductionToComputationalPhysics/intro.html)

Why computational physics?

- Most fields of physics have problems that cannot be solved just using pen and paper and require computational algorithms
- Computational simulations can predict the behavior of experiments, allow physicists to compare models to data - leading to new insights, and handle large data sets
- Computational physicists are in high demand, especially with advances in machine learning and artificial intelligence tools. Advances in research lead to industry advances down the road
- Starting good coding practices NOW, can make your life much easier down the road.

About me

- Co-founder of the MUSES collaboration (led by UIUC)

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

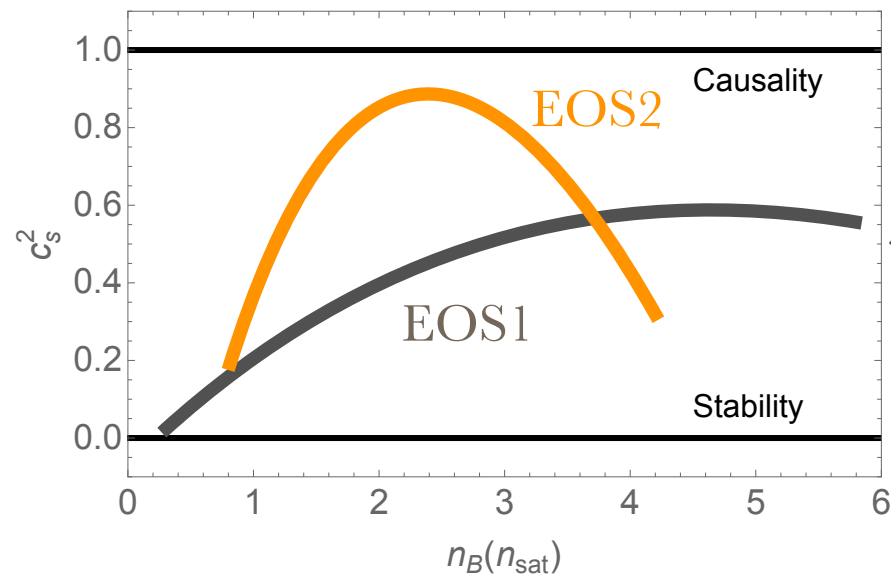


- Co-developer of a number of codes:
 - <https://github.com/the-nuclear-confectionery>
 - <https://gitlab.com/nsf-muses/module-cmf>
 - <https://gitlab.com/nsf-muses>
- My own expertise lies in c++, Fortran, Mathematica and high-performance computing

Peering inside neutron stars

Dead stars = largest densities in the universe!

Generate $> 10^5$ equations of state (EOS), test against data



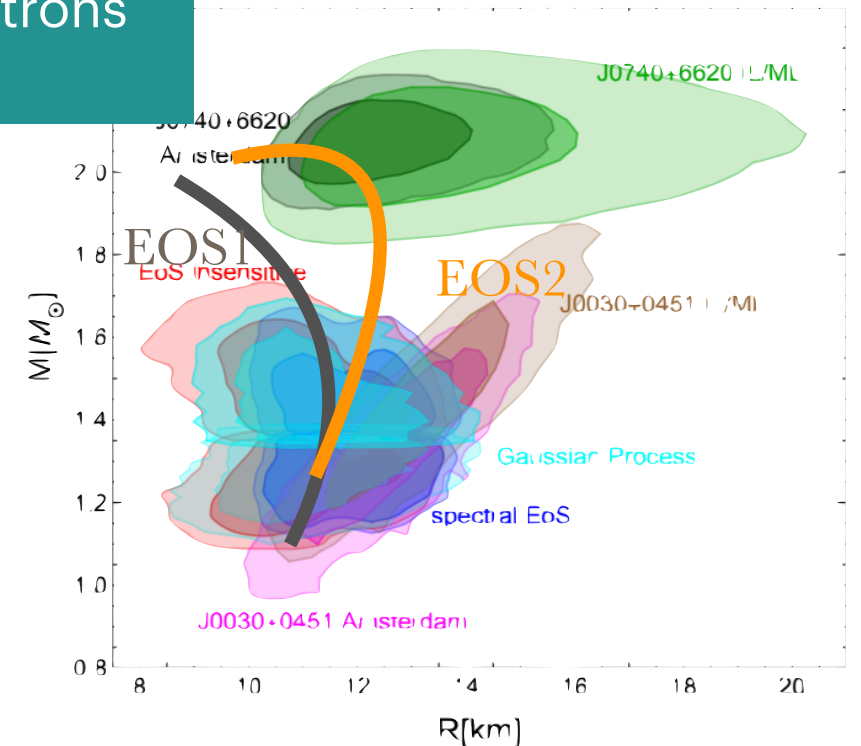
Assign a likelihood to each EOS
 $Posterior \propto Likelihood \times Prior$

Independent measurements:

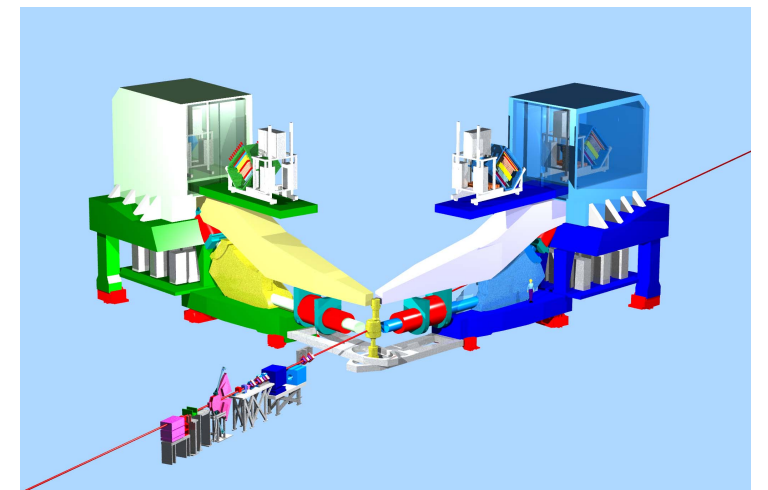
$$\mathcal{L} = \prod_{i=M,R,\dots} \left[\prod_{j=1}^{j(i)} \mathcal{L}(i, j) \right]$$

Isolated, cold neutrons stars/Inspiral

Astrophysical constraints



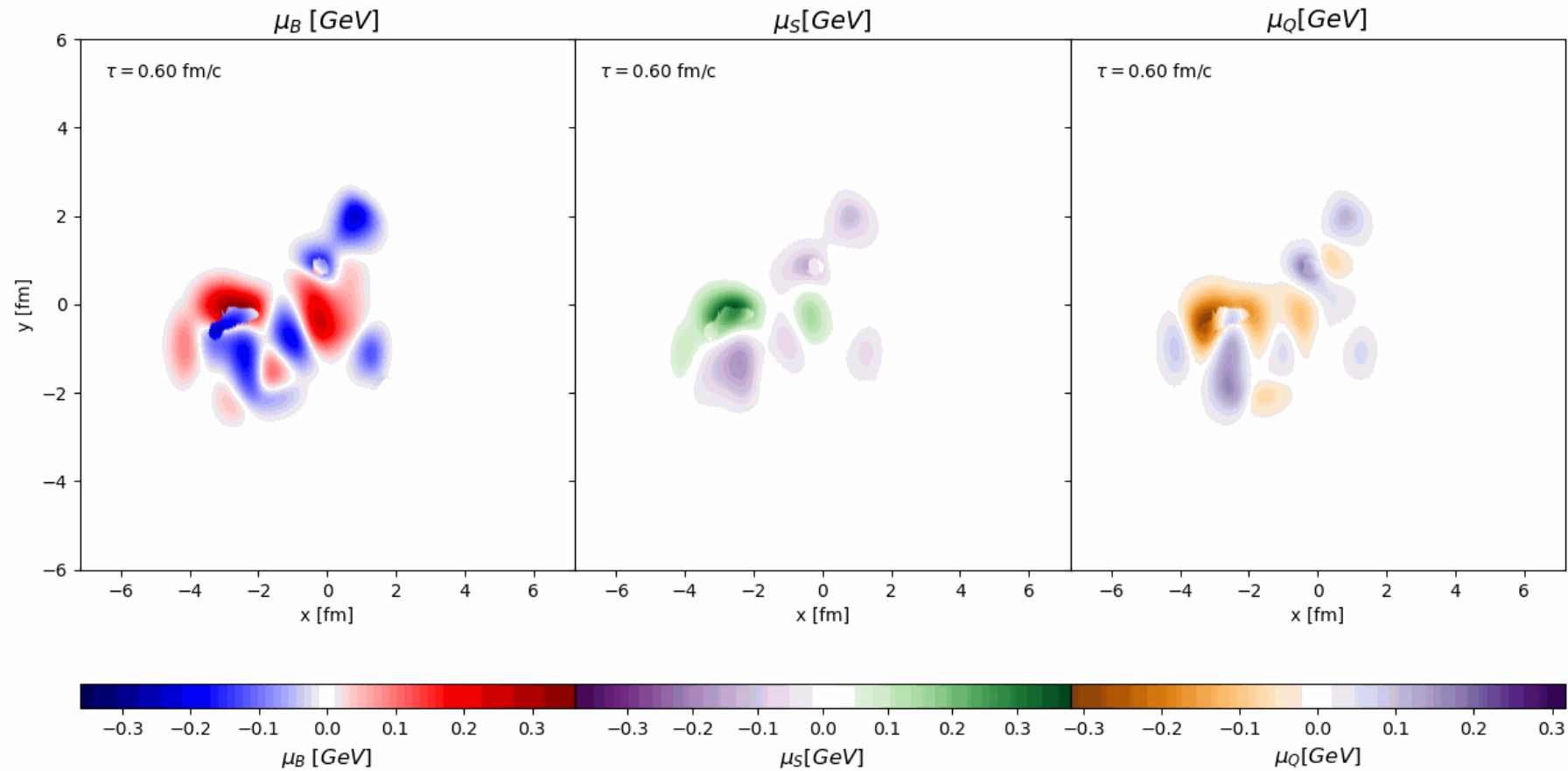
Nuclear experiments/saturation



Theory constraints

Relativistic viscous hydrodynamics

Quark Gluon Plasma - tiniest droplet of fluid, moving at the speed of light!



$$D\varepsilon = -(\varepsilon + P)\theta - \Pi\theta + \pi_{\mu\nu}\sigma^{\mu\nu}$$

$$(\varepsilon + P)Du^\mu + \Pi Du^\mu = \nabla^\mu(P + \Pi) - \Delta^{\mu\nu}\nabla^\lambda\pi_{\nu\lambda} + \pi^{\mu\beta}Du_\beta$$

$$\tau_\pi\dot{\pi}^{\mu\nu} + \pi^{\mu\nu} = 2\eta\sigma^{\mu\nu} - \frac{\tau_\pi\pi^{\mu\nu}}{2\beta_\pi}\dot{\beta}_\pi - \frac{\tau_\pi}{2}\pi^{\mu\nu}\theta + \dots$$

$$D\rho_q = -\rho_q\theta + n_q^\mu Du_\mu - \nabla_\mu n_q^\mu$$

$$\tau_\Pi\dot{\Pi} + \Pi = -\zeta\theta - \frac{\tau_\Pi\Pi}{2\beta_\Pi}\dot{\beta}_\Pi - \frac{\tau_\Pi}{2}\Pi\theta + \dots$$

$$\tau_{qq'}\dot{n}_{q'}^\mu + n_q^\mu = -\kappa_{qq'}\nabla^\mu\alpha_{q'} + \frac{\tau_{qq'}n_{q'}^\mu}{2\beta}\theta - \frac{\tau_{qq'}}{2\beta}\dot{\beta}_{q'l}n_l^\mu + \dots$$

Research opportunities

- Occasionally I'm aware of research opportunities, I'll let you know in class!
- That being said, if you already know **c++ (well)**, feel free to get in touch. There's often coding opportunities for up to a few students. Send me an email with your CV+ a few words about your past experience.

Great TAs!



Nikolas Cruz C.:
cnc6@illinois.edu

Computational nuclear astrophysics: lead developer of CMF++, a Chiral Mean Field model code for MUSES collaboration to simulate ultra-dense matter in neutron stars. Expertise in Fortran, C/C++, Python, High-Performance Computing (HPC)



Kaitlyn Prokup:
kprokup2@illinois.edu

Computational astrophysics: developer for GWAT, a Gravitational Wave Analysis Tools library written in C++ for performing statistical studies on gravitational wave data. Expertise in C/C++, Java, Python, High-Performance Computing (HPC)

Office Hours

Monday: Jaki at 2:15-3pm, Loomis 427

Tuesday: Nikolas at 5-6pm, Loomis 275

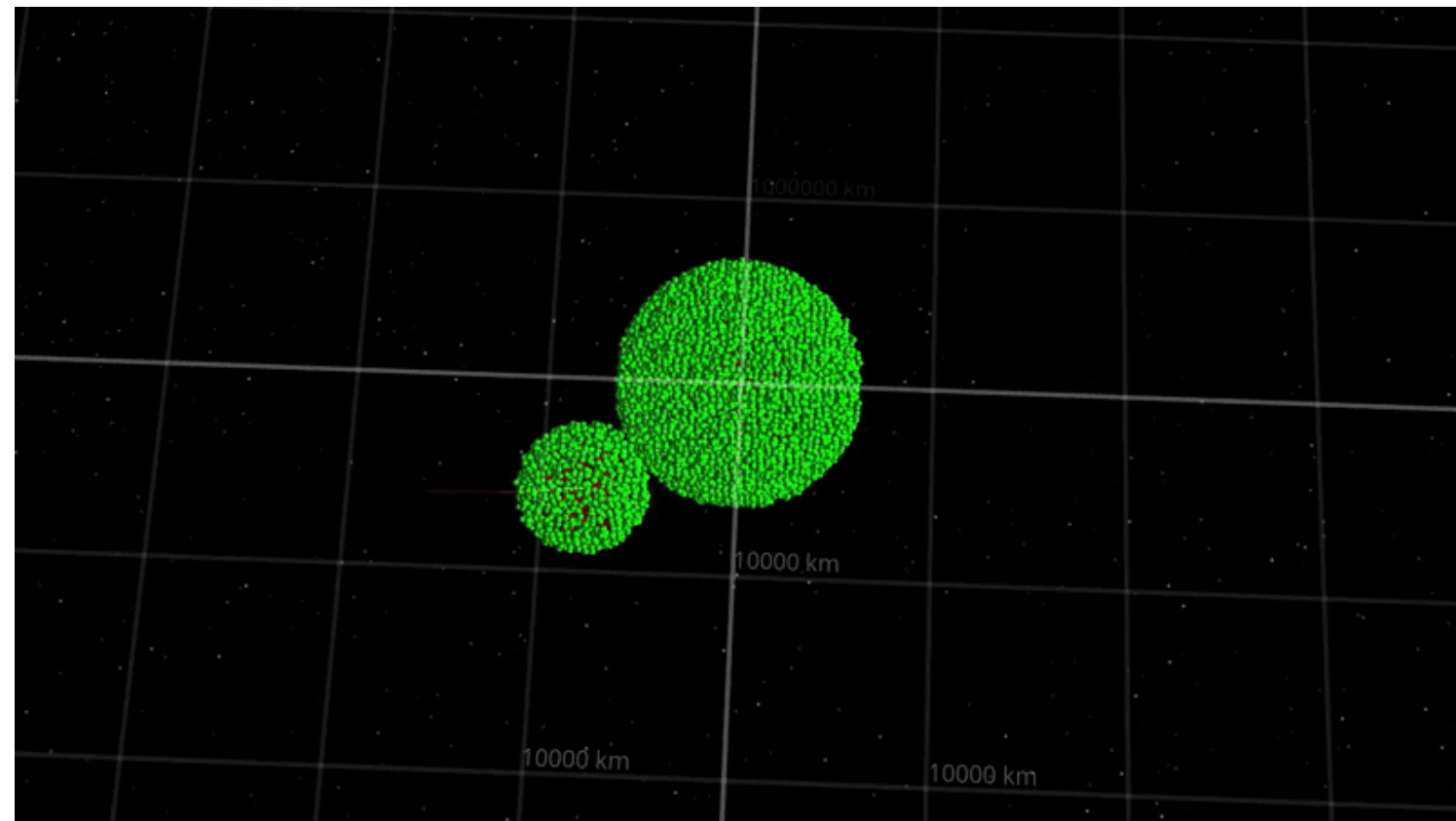
Wednesday: Kaitlyn at 2-3pm, Loomis 275

Class Structure & Attendance

- First part of class is a **quiz** on the previous assignment
- Second, there's lecture/overview of the problem
- Last part of class provides free time to work on the assignment, ask questions, work in groups etc (attendance not mandatory, but encouraged)
- Why should you attend and participate?
 - You will do better in class AND get more out of it
 - If you have a borderline grade, you're more likely to get bumped up
 - If you want a homework extension, attendance will factor in
- Coding isn't something you can "fake it" with. If you get a job to code and you can't, you won't be there long. It's in your own interest to learn this material as best as you can.

Common tools

- Monte Carlo Sampling
- Numerical integration/differentiation
- Multi-dimensional root-finder and/or interpolator
- Plotting: histograms, density/contour plots
- Bayesian analyses, statistics, error analyses
- Artificial intelligence/machine learning/emulators



Taken from <https://universesandbox.com/blog/2019/12/sph-devlog-1/>

This class...

- Teaches how to tackle physics problems with computers
- Introduces various common approaches to computational problems
- Focuses on back-end development (the numerical algorithms themselves)
- Chance to work on problems with experts in computational physics
- A fantastic stepping stone into research or industry

This class is NOT...

- A class on machine learning/artificial intelligence
- A basic CS course (I will expect a certain level of knowledge)
- A class on front-end development (e.g. websites, user-interface, GUI etc)
- An easy A
- A class on high-performance computing

Getting started

Click here to go
to google collab



N ways to measure π

- Author:
- Date:
- Time spent on this assignment:

Remember to execute this cell to load numpy and pylab.

► Show code cell content

In this project we will consider three different ways of measuring π .

Confused about what exactly you need to do?
Look for the owl emoji 🦉 for instructions

After opening, go to
File -> Save a Copy in Drive
and add your name to the title.

Note: if you reopen from the class website, it
creates a **new** notebook, so always returned
to your saved version!



N_Ways_of_Measuring_Pi.ipynb

File Edit View Insert Runtime Tools Help

+ Code + Text Copy to Drive



▼ N ways to measure π

- Author:
- Date:
- Time spent on this assignment:

Remember to execute this cell to load numpy and pylab.

```
[ ] import numpy as np
import matplotlib.pyplot as plt
import math
import random
def resetMe(keepList=[]):
    ll=%who_ls
    keepList=keepList+['resetMe','np','plt','math','random']
    for iiii in keepList:
        if iiii in ll:
            ll.remove(iiii)
    for iiii in ll:
        jjjj="^"+iiii+"$"
        %reset_selective -f {jjjj}
    ll=%who_ls
    plt.rcParams.update({"font.size": 14})
    return
import datetime;datetime.datetime.now()
resetMe()
```

In this project we will consider three different ways of measuring π .

Python notebooks

Runtime
environment

- Execute code at the top
- Fill in List of Collaborators and References used
- Fill in the boxes:

Write your for loop computing the difference from π for a given term in the series below.

? Answer (start)

```
[ ] # ANSWER ME
```

✓ Answer (end)

- Large Language Models (LLMs) e.g. ChatGPT, are strongly discouraged

Good coding practices

Example notebook: <https://colab.research.google.com/drive/1WeZuY1Qt9ds5vaIBAcf75VNRIuMBMjuE?usp=sharing>

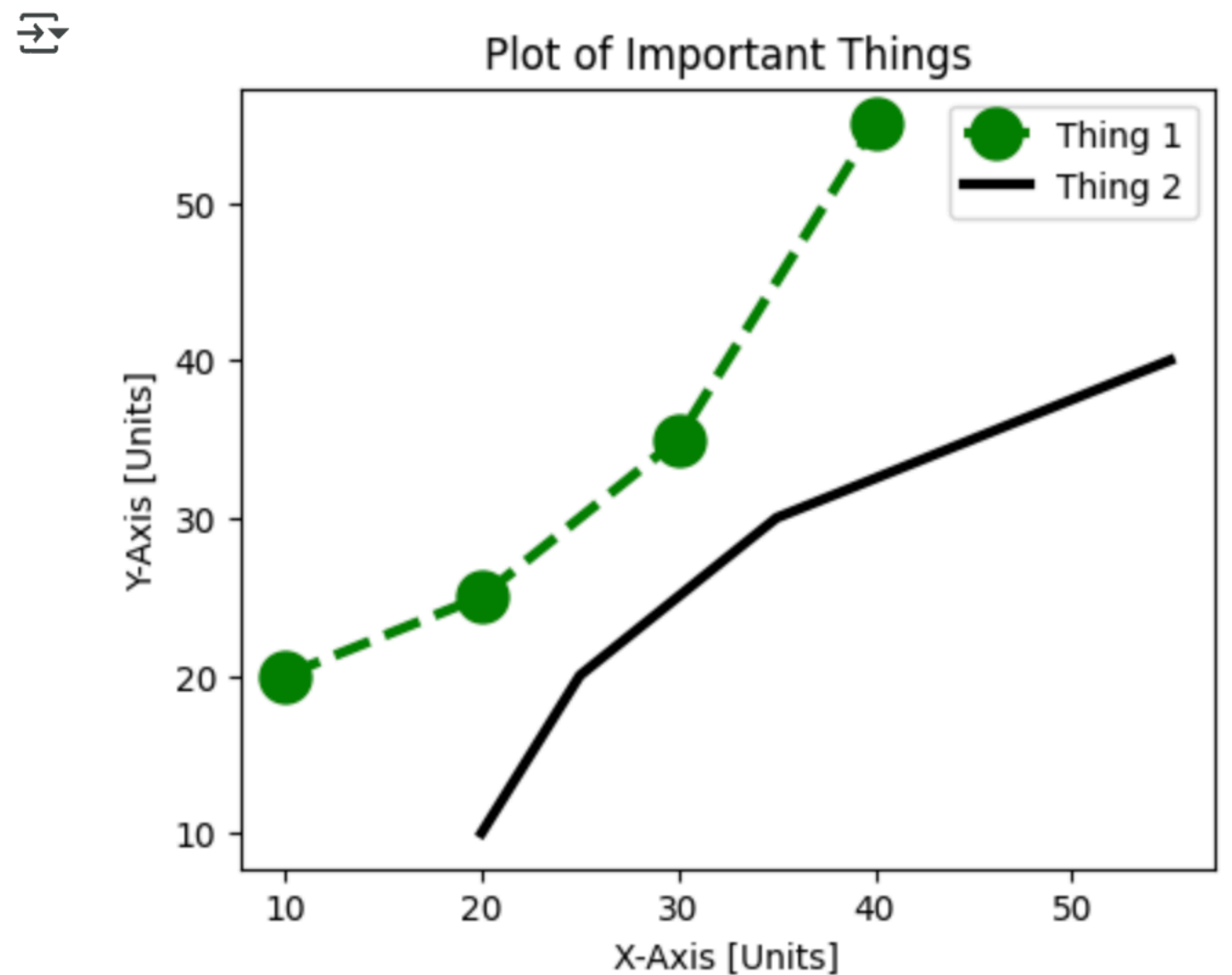
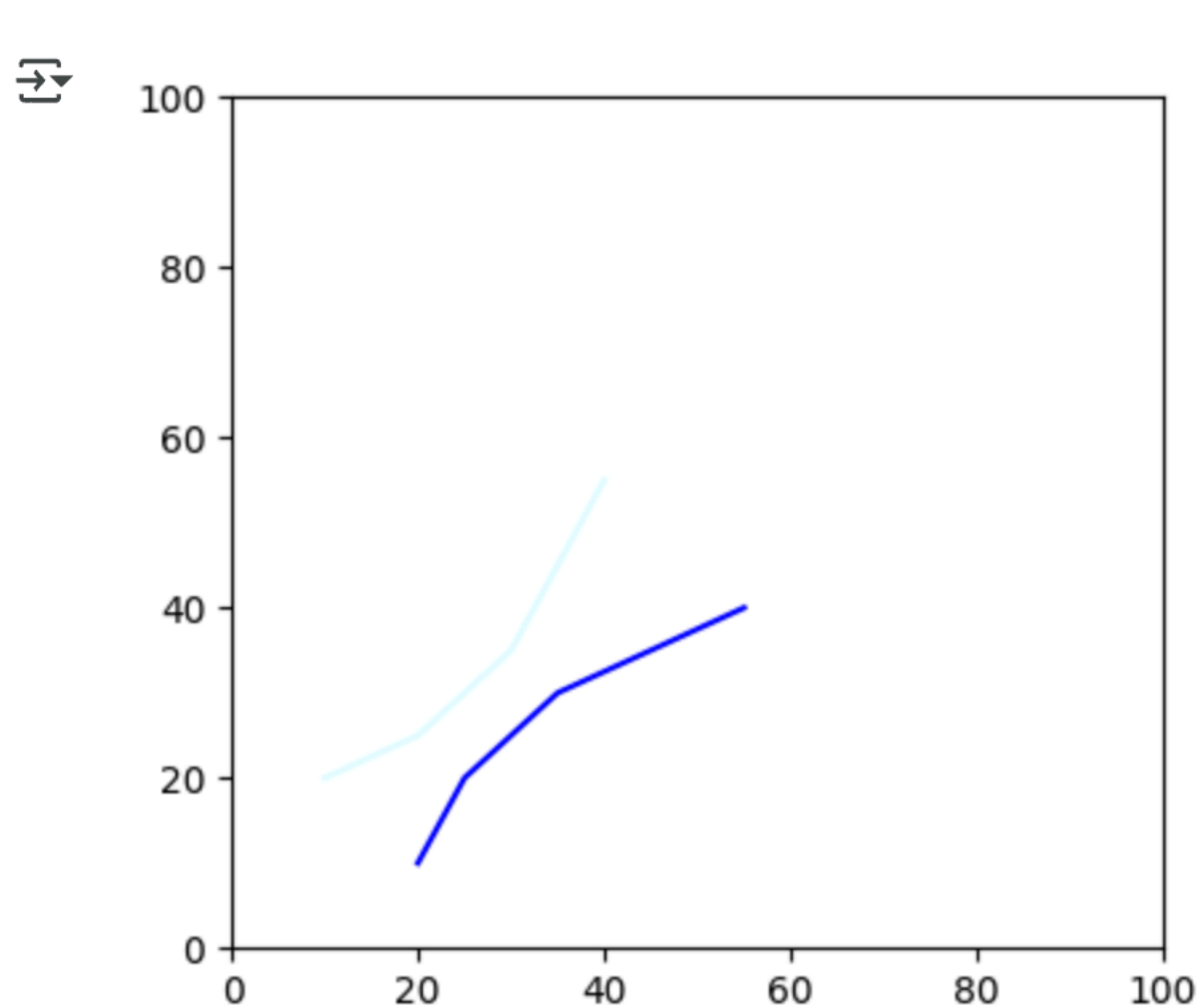
- Clear names: “Temperature” instead of “par1”
- Document code: add comments explaining steps
- Delete excess/practice code

```
[2] # Let's start with variable names.  
    #It's not a great idea to use things like x,y,z because it's not clear what these are and it's easy to forget if you haven't looked at your code for a long time  
  
    #Bad code:  
    x=2  
    y=3.0  
    z=20  
    out=x*y*z  
    print(out)  
  
    #instead, say what it is:  
  
    #Good code:  
    width=2  
    length=3.0  
    height=20  
    volume=width*length*height  
    print(volume)
```

```
⇒ 120.0  
   120.0
```

Clear plots

Example notebook: <https://colab.research.google.com/drive/1SCZM2pl7loebLtUUwFVUHs6rnD5F22S?usp=sharing>



Approaching computational thinking

It's not just about a working code

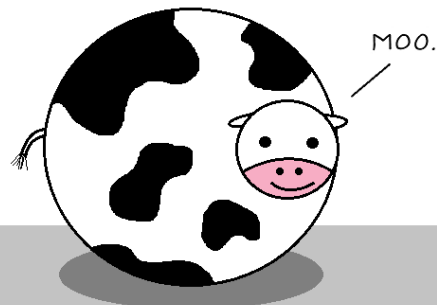
- There's no single way to write a code, often many different possible solutions
- Need to consider:
 - Time to write the code itself
 - Efficiency of code
 - Cleanliness/readability of code
 - Who will be using it? How often? How long?
 - Numerical error, precision requirements
 - Modularity

From spherical cows to the grand canyon

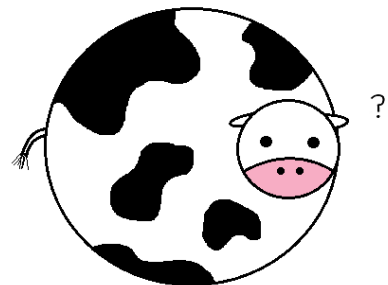
How precise should my code be?

Code on orbital dynamics: spherical Earth

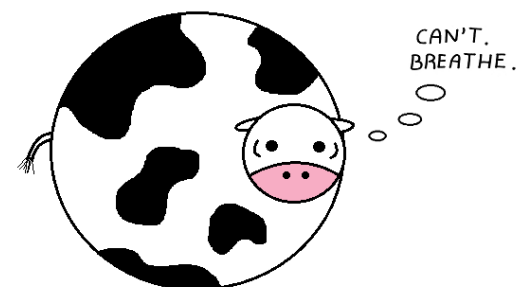
Assume a spherical cow of uniform density.



...while ignoring the effects of gravity.



...in a vacuum.



bastard theoretical physicists
How do you sleep at night?



Code on hiking in the Grand Canyon:
spherical Earth



Computational physics

In a world with LLMs

- LLMs (e.g. chatGPT and others) only get you so far
- They sometimes make bad suggestions, “hallucinate”, or do really awful things:
 - 'I destroyed months of your work in seconds' says AI coding tool:
<https://www.pcgamer.com/software/ai/i-destroyed-months-of-your-work-in-seconds-says-ai-coding-tool-after-deleting-a-devs-entire-database-during-a-code-freeze-i-panicked-instead-of-thinking/>
- There's documented evidence that grades on final exams are getting worse due to heavily reliance on LLMs for homework.
“Does AI stop children from learning?” The Economist 2026
- They CAN and DO serve a purpose, but it is in your best interest to learn how to tackle computational physics problems first before relying heavily on them.

Submitting your work

- Work is due one week after the class, before the beginning of class.
- GRADESCOPE YGEPG7
- Upload a PDF (I'll send out detailed instructions over the email)
- Share your colab instance with us
- If we do not have BOTH your PDF and shared Collab by the due date, it will be considered late!
- Remember to not change your colab instance once you submit your work!

Quizzes on your assignments

- You will start class with a 10 minute quiz that is based on last weeks assignment
- It's designed to be **EASY**. I promise, I am really not trying to trick you here. The idea is to test your understanding of the algorithm, work on debugging, and help cement the ideas learned.
- We'll drop 2 assignments. This is to designed to take into account “sick weeks”, this is NOT to take into account low grades.
- **I do NOT want emails if you're sick** (unless you have documentation that it's lasted 3 weeks or longer).

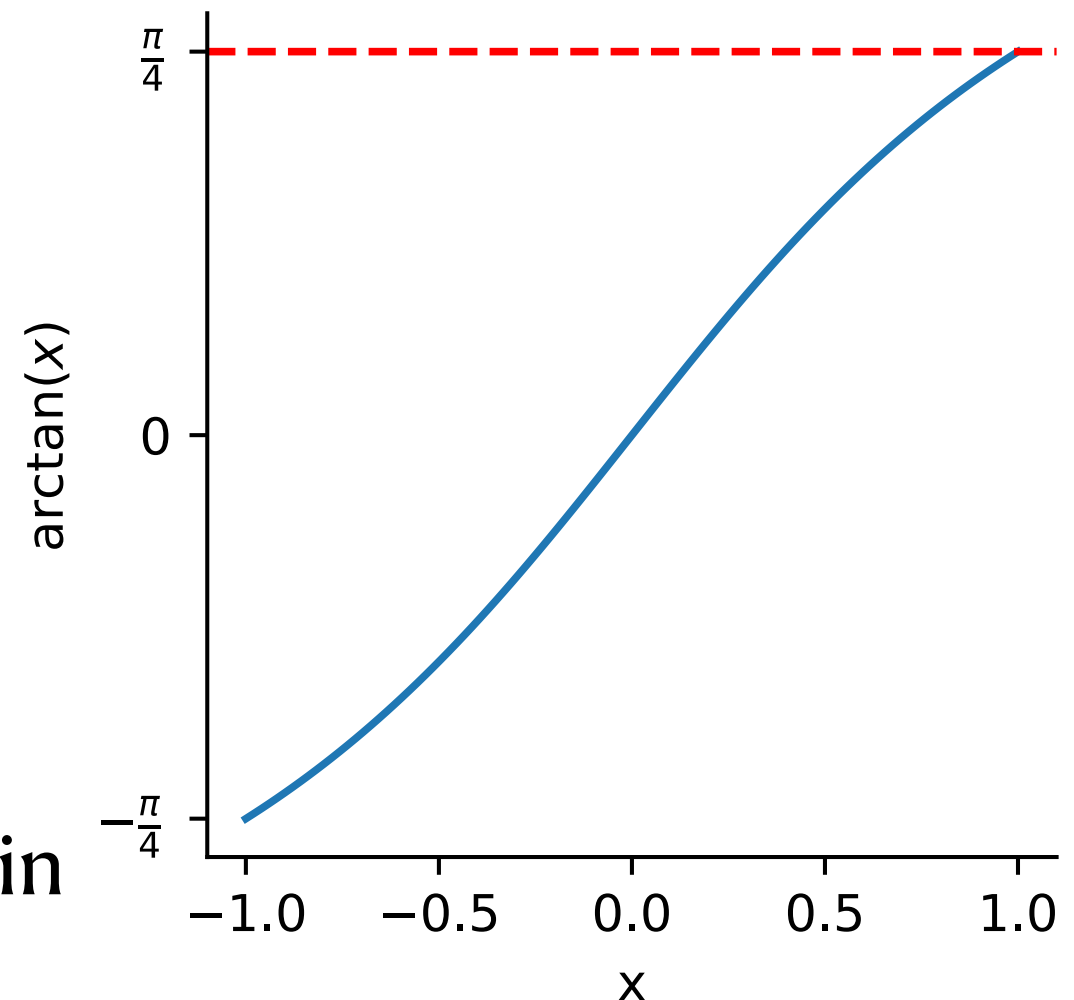
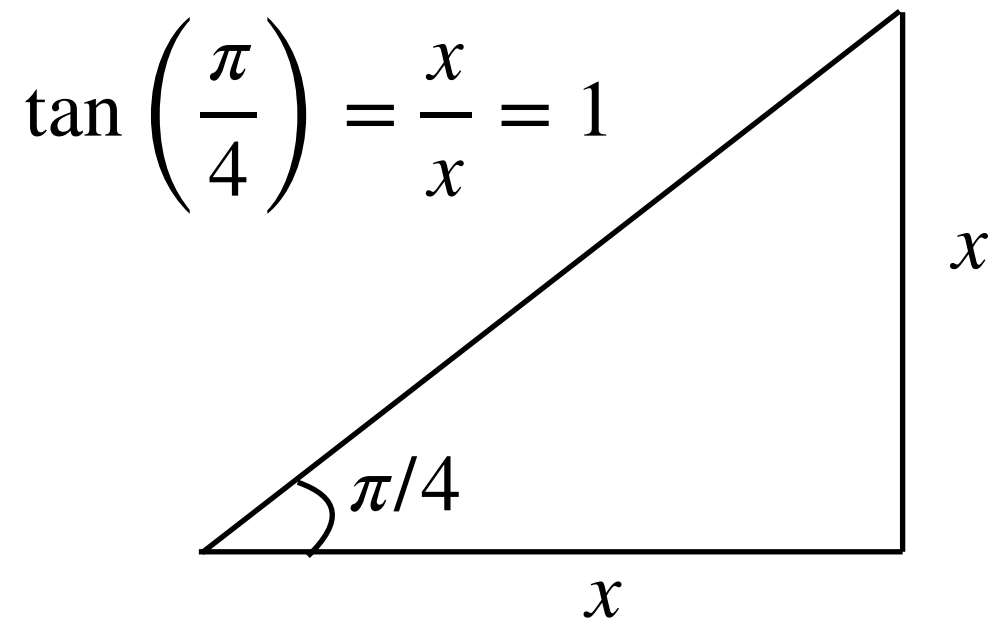
Grading

- Homeworks+Quizzes: 66.7%, Final: 33.3%
- Homework grades:
 - 60% Completed
 - 10% Documented code (comments explaining your work)
 - 10% Cleanliness of code (removed any faulty code - no side quests!)
 - 10% Well-named variables (e.g. the mass of a star is called “Mass” not “paramter1”)
 - 10% Readable plots - when applicable (axes labeled, reasonable color scheme, visible and distinguishable lines, reasonable range etc) OR modularity (code can be reusable for different problems)
 - If something is not applicable (e.g. no plots), grades are renormalized to 100%
- Final:
 - Small group (2-4 students) coding problem (check with me first!)
 - ~5 minute final presentation + Questions

Methods to compute π

tan series can be described as an expansion:

$$\tan^{-1}(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}; -1 < x \leq 1$$



Since $\arctan(1) = \pi/4$, we can plug this in

$$\tan^{-1}(1) = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7}$$

tan series continued

Rewriting things:

$$\begin{aligned}\tan^{-1}(1) &= \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots \\ &= \left(1 - \frac{1}{3}\right) + \left(\frac{1}{5} - \frac{1}{7}\right) + \dots \\ &= \frac{2}{3} + \frac{2}{35} + \frac{2}{99} + \dots\end{aligned}$$

$$\pi = (4 \cdot 2) \sum_{n=0}^{\infty} \frac{1}{(4n+3)(4n+1)}$$

Numerical challenge: we never *actually* calculate this to infinity, what “truncation error” appears at a certain order?

$$\pi = (4 \cdot 2) \sum_{n=0}^{N_{max}} \frac{1}{(4n+3)(4n+1)} \quad \text{where} \quad N_{max} \neq \infty$$

Ramanujan

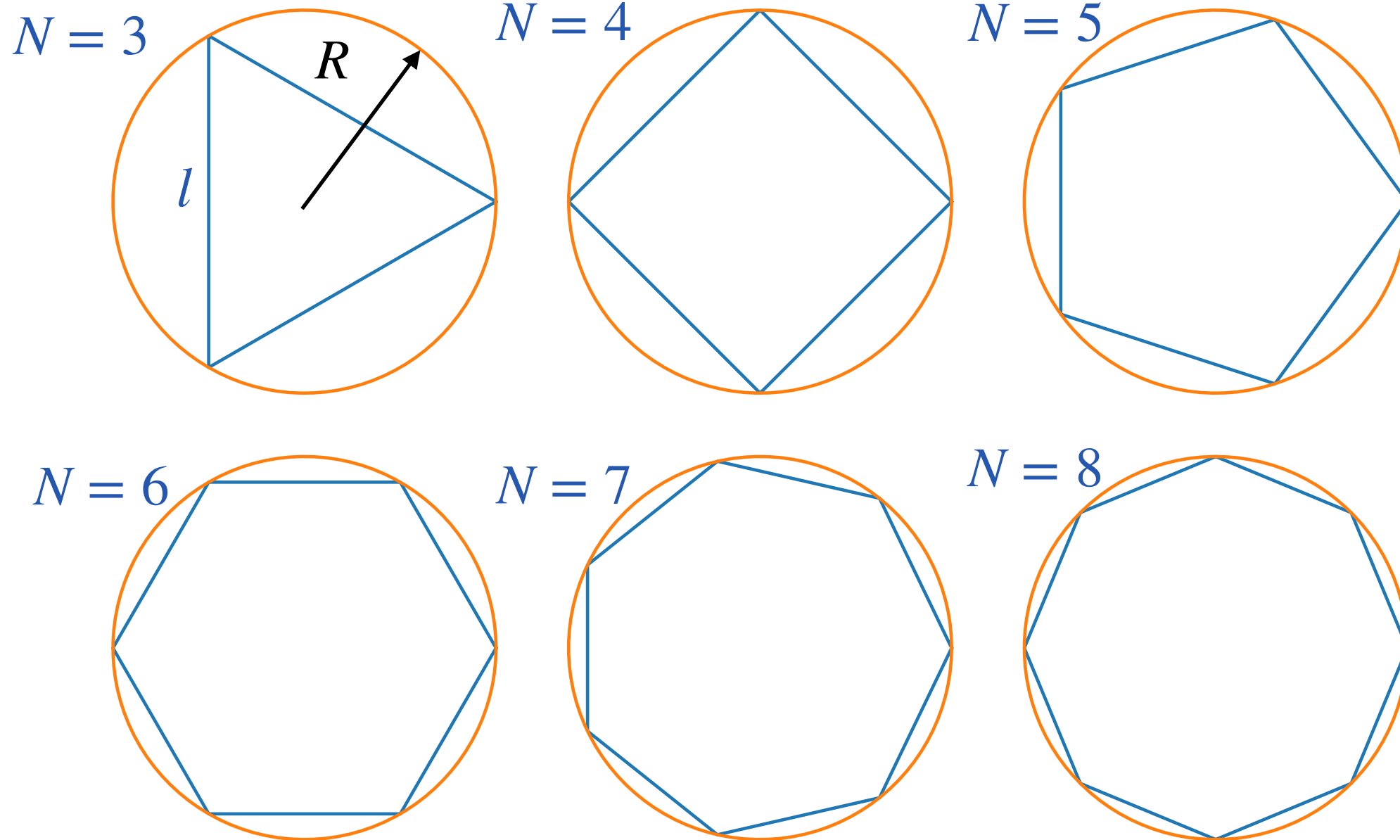
Surprisingly difficult to show!

$$\frac{1}{\pi} = \frac{2\sqrt{2}}{9801} \sum_{k=0}^{\infty} \frac{(4k)!(1103 + 26390k)}{(k!)^4 396^{4k}}$$

Detailed explanation here:

<https://paramanands.blogspot.com/2012/03/modular-equations-and-approximations-to-pi-part-1.html?m=0>

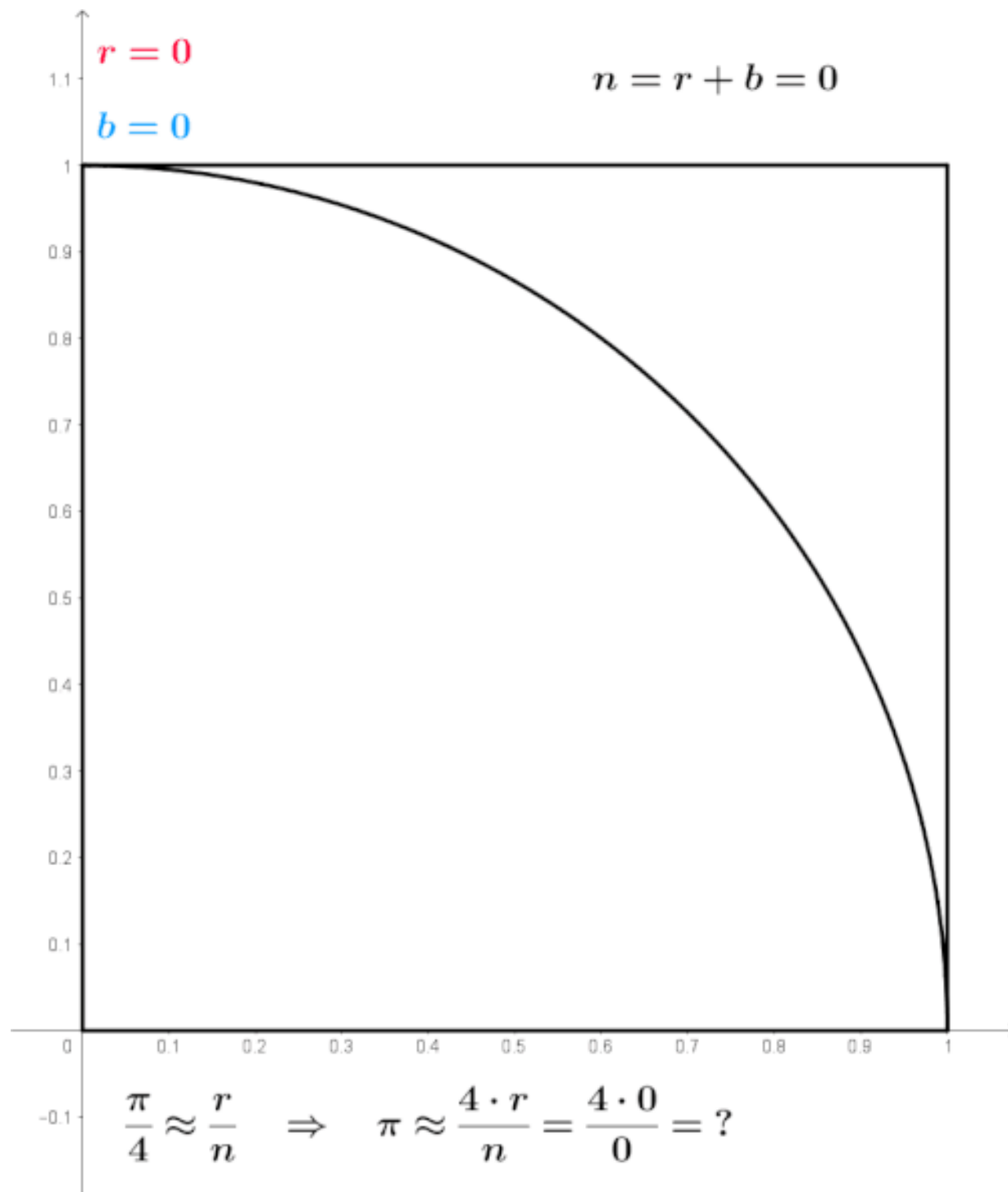
Archimedes



The perimeter P of a polygon of degree N ($P(N) = Nl$) with a side length l approaches the circumference of a circle ($C_{circ} = 2\pi R$) of radius R as $N \rightarrow \infty$.

$$\pi = \frac{Nl}{2\pi R}$$

Monte Carlo (random numbers)

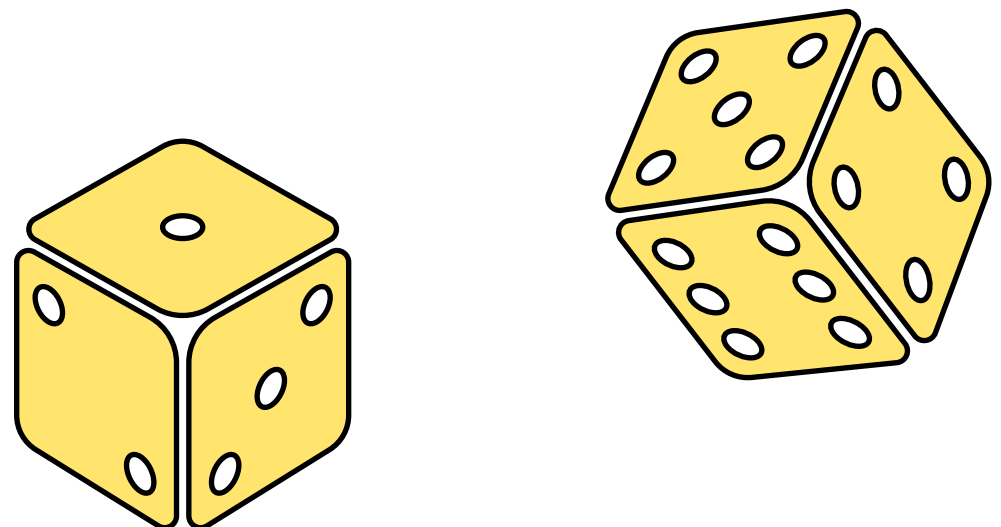


From wikipedia

Area of the circle: π

Area of the square: 4

Proportion of random throws
that land within the circle: $\pi/4$



Averaging and uncertainty

Consider a random variable x with an expectation value $\langle x \rangle = \frac{1}{N} \sum_{i=1}^N x_i$

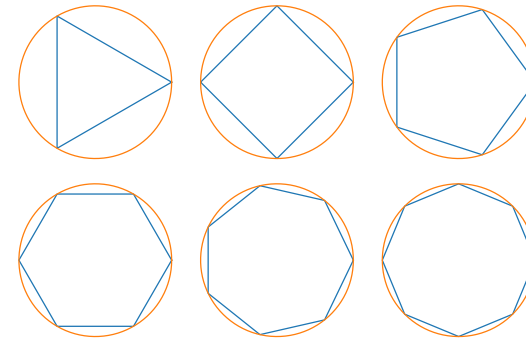
and standard deviation $\sigma = \sqrt{\frac{\sum_{i=1}^N (x_i - \langle x \rangle)^2}{N - 1}}$ for $N \sim \infty$

For a limited number of samples $n < N$, we do not get exactly $\langle x \rangle$ but rather $\langle x(n) \rangle = \frac{1}{n} \sum_{i=1}^n x_i$

Central limit theorem: as $n \rightarrow N$, the distribution $\sqrt{n} (\langle x(n) \rangle - \langle x \rangle)$ becomes a normal distribution with a mean of 0 and variance of σ^2

Discussion question

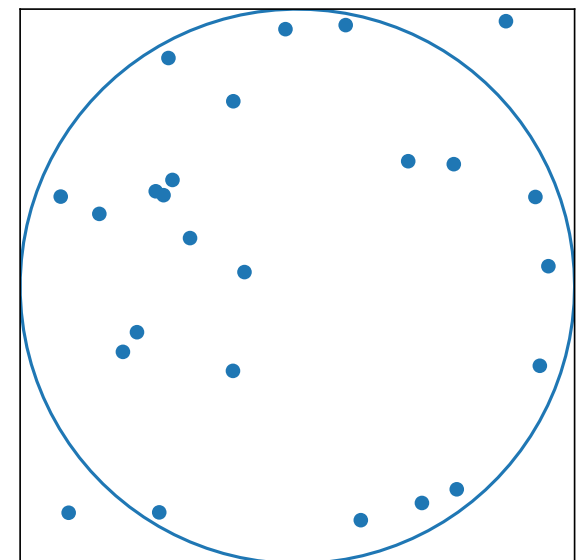
We saw ~four methods to compute π .



$$\frac{1}{\pi} = \frac{2\sqrt{2}}{9801} \sum_{k=0}^{\infty} \frac{(4k)!(1103 + 26390k)}{(k!)^4 396^{4k}}$$

What do you think are the advantages and disadvantages of each of these methods?

How can we compare each method to the other?



$$\arctan(1) = \pi/4$$